Fiscal Year:	FY 2017	**Task Last Updated:**	FY 05/14/2018
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Skeletal Responses to Long-Duration Simulated Microgravity in Male and Female Rats		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Ruth.K.Globus@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	08/01/2015	**End Date:**	09/30/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	Biospecimen Sharing NOTE: Extended to 9/30/2018 (original end date was 6/30/2017) per PI/ARC (Ed., 5/8/18) NOTE: Extended to 9/30/2017 (original end date was 6/30/2017) per A. Ronca/ARC (Ed., 5/4/17) NOTE: End date should be 6/30/2017, per A. Ronca/NASA ARC (Ed., 5/11/16) NOTE: Extended to 12/1/2016, per A. Ronca/NASA ARC (Ed., 5/10/16) NOTE: change in period of performance to 8/1/2015-7/31/2016, per A. Ronca/NASA Ames (original period of performance was 6/25/14-9/30/16)--Ed., 3/27/15		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Blaber, Elizabeth Ph.D. (NASA Ames Research Center) Shirazi-Fard, Yasaman Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Long duration spaceflight can cause severe bone loss, particularly at the lumbar spine and hip. The rate of bone loss in crewmembers while on the space station can be profound, and a significant increase in estimated fracture risk remains even one year after returning to Earth. In addition to microgravity, the natural aging process leads to declines in bone mass and also increases in fracture risk, which suggests that the combination of the two could have compounding effects on bone structure and integrity. Changes in bone structure and strength have unknown consequences and important implications for the skeletal health of crewmembers. In recent International Space Station (ISS) missions, crewmembers have maintained bone mass better, presumably due to improved resistance exercise protocols coupled with adequate energy intake and vitamin D; however, no predictors of bone geometry or strength were evaluated. As NASA is preparing to send astronauts to the ISS and beyond for missions exceeding one year, it is critical to characterize and understand bone loss and recovery for longer duration spaceflight. The proposed project will take advantage of the unique tissue sharing opportunity for obtaining frozen specimens from a University of California (UC) Davis experiment that entails 90-day rat unloading and reloading studies. To accomplish our Aims, we will determine how responses to simulated weightlessness vary in young (3 month old) vs. adult (9 month old) male rats as a function of time, using 3D imaging (microcomputed tomography), mechanical testing, and molecular gene expression analyses. We also will determine if bone loss and recovery differ between male and female rats. This project addresses risks and fills gaps highlighted in NASA's Human Research Program that focus on better understanding risk and preventing space-related bone loss.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from this research will identify basic molecular mechanisms that may underlie degenerative skeletal changes that arise due to prolonged musculoskeletal disuse on Earth. The findings are relevant to osteoporosis, a common and often devastating skeletal diseases. On Earth, osteoporosis can result from aging, hormonal changes, immobilization, prolonged inactivity, and bedrest. In addition, osteoporosis affects the ability of skeletal tissue to recover from fractures. Identification of specific mechanisms that contribute to the initiation and progression of osteoporosis with the hindlimb unloading model of spaceflight will contribute to the development of new countermeasures.
Task Progress:	Spaceflight can profoundly affect skeletal structure and the expression levels of genes related to oxidative stress. However, the molecular links between oxidative stress, bone loss, and weightlessness in skeletal tissue are not yet fully understood. We hypothesized that gene expression changes in oxidative defense-related pathways active processes involved in bone breakdown precede changes in skeletal structure during long duration simulated weightlessness and recovery. To test our hypothesis, we are performing 3-dimensional imaging by microcomputed tomography, biomechanical strength testing, and gene expression pathway analysis in the long bones from the hindlimbs of adult male and female rats that underwent various periods of simulated weightlessness by hindlimb unloading (HU, up to 90 days) and re-ambulation (up to 90 days). Preliminary microcomputed tomography (microCT) results from the femoral midshaft revealed that 90 days of simulated weightlessess in males does lead to bone loss although 14 days has no effect. Evidence of detailed changes in bone geometry suggest some compensatory process occurred during the lengthy experimental period. We also have purified RNA from samples and find that the majority of samples are of sufficiently high quality for detailed analysis of all the RNA transcripts expressed in the bone tissue (RNAseq).
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Abstracts for Journals and Proceedings	Globus R. "Interactive Effects of Microgravity and Radiation: Effects of Biological Sex and Age." President's Plenary : New Approaches to Translational Space Medicine. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Rael VE, Shirazi-Fard Y, Tahimic C, Globus RK. "Gene Expression and Structural Skeletal Responses to Long-Duration Simulated Microgravity in Rats." Poster session. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. 32nd Annual Meeting of the American Society for Gravitational and Space Research, Cleveland, OH, October 26-29, 2016. , Oct-2016
Abstracts for Journals and Proceedings	Shirazi-Fard Y, Rael V, Torres S, Bryant S, Tahimic C, Globus RK. "Gene Expression and Structural Skeletal Responses to Long-Duration Simulated Microgravity in Rats." Poster session. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. 2017 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 23-26, 2017. , Jan-2017
Articles in Peer-reviewed Journals	Tahimic CGT, Globus RK. "Redox signaling and its impact on skeletal and vascular responses to spaceflight." Int J Mol Sci. 2017 Oct 16;18(10):E2153. Review. https://doi.org/10.3390/ijms18102153 ; PubMed PMID: 29035346; PubMed Central PMCID: PMC5666834 , Oct-2017